INCH-POUND

MIL-S-70791 (AR)
9 April 1990

MILITARY SPECIFICATION

SIMULATOR, MISSILE: SAGGER XM27, PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts for and loading, assembling and packing for one type of simulator designated as Simulator, Missile: sagger XM27.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1370

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SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation Methods of
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For.
MIL-P-70762	-	Propellant, Composite, Cast, For use in Simulator, Missile: sagger XM27

STANDARDS

MILITARY

MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests For

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

12576572	-	Simulator, Missile: sagger XM27
12598324	-	Packing and Marking For Support, Shipping, Polystyrene Foam For Simulator, Missile: sagger XM27
12598333	-	Packing and Marking for Box, Pallet Type, Wirebound for Simulator, Missile sagger XM27

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(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings, specifications and standards.

3.2 Assembly. The Simulator Assembly shall comply with all requirements specified on Drawing 12576572 and all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Propellant. The propellant shall comply with all requirements of MIL-P-70762 and all applicable drawings.

3.3.1 Loaded rocket. Total area of cracks and voids on the propellant surface shall not exceed 2 square inches.

3.3.1.1 Rocket propellant adhesion. Propellant shall exhibit no visual signs of separation from the tube wall.

3.4 Function. The simulator shall comply with the following requirements when fired using igniter assembly (dwg. 9390399), an electric match and twelve (12) volt power source, from an approved fixture capable of firing up to an angle of 30 degrees maximum (max) from the horizontal.

a. The rocket shall expel a visible flame and minimal light grey plume, both visible during powered flight.

b. The rocket shall not impact the ground for the first 1000 feet of flight.

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c. The rocket range shall be a minimum of 1000 ft.

3.5 Environmental Conditions.

3.5.1 Rocket assembly integrity test (Dwg. 12576571). The nose and tailcone shall not separate from the rocket tube and shall not exhibit any signs of bulging or displacement when tested as specified in 4.5.5.

3.5.1.1 Dimensional stability test. The rocket assembly shall exhibit dimensional stability (less than 2 percent variation) when tested as specified in 4.5.5.1.

3.5.2 Transportation vibration - temperature. The rocket simulator (Dwg. 12576572) shall comply with the following requirements when tested as specified in 4.5.3.

a. There shall be no loose propellant on the simulator or in the test box after testing.

b. The rocket components shall not separate during test.

c. The rocket shall not function during test.

3.5.3 Five foot drop test. The loaded rocket (Dwg. 12576572) shall not function and shall remain fully operable when tested as specified in 4.5.4 and 4.5.2.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, blisters, excess flash, scratches, dirt, grease and other defects and foreign matter which would affect appearance or serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined. All packing components shall be dry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual nor does it commit the Government to acceptance of any defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items and sample quantities as indicated in Table I:

Part Description	Drawings
Nose	12576516
Tube, Rocket	12576517
Tailcone	12576518
Simulator, Missile: sagger XM27	12576572
Box, Pallet Type, Wirebound	12597833
Support	12598581

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4.3.2 Inspections to be performed. First article assemblies, components and test specimens shall be subjected by the Government to any or all of the examinations and tests specified in Table I and MIL-A-48078.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-S-70791 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Simulator, Missile: sagger XM27		See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Tube, Rocket (Dwg. 12576517)			
	Examination for defects	10	3.2	4.4.2.2
	Nose (Dwg. 12576516)			
	Examination for defects	3/cavity	3.2	4.4.2.1
	Tailcone (Dwg. 12576518)			
	Examination for defects	3/cavity	3.2	4.4.2.3
	Rocket Assembly (Dwg. 12576571)			
	Integrity test	30	3.5.1	4.5.5/Test
	Dimensional stability	30	3.5.1.1	4.5.5.1

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-S-70791 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
	Simulator, Missile: sagger XM27		See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Simulator, Missile: sagger XM27 (Dwg. 12576572)			
	Examination for defects	50	3.2	4.4.2.6
	5 ft. drop	10(a)	3.5.3	4.5.4, 4.5.2
	Function			
	Hot	10(a)	3.4	4.5.2
	Cold	10(a)	3.4	4.5.2
	Ambient	10(a)	3.4	4.5.2
	Transportation Vibration - Temperature	10(a)	3.5.2	4.5.3, 4.5.2

NOTES:

(a) Samples for this test shall be taken from the above 50.

AMSMC Form 1570b, 1 Jul 89

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Simulator, Missile: sagger XM27 assemblies shall contain.

a. Parts of one kind from one manufacturer.

b. Propellant ingredients from not more than one lot or batch (applicable only when not manufactured by the loading contractor).

4.4.2 Examinations and tests.

a. Classification of Characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein.

Sampling shall be in accordance with the following Table II using the inspection levels specified therein. If one or more defects are found in any sample, the lot shall be rejected.

Table II. Sampling Inspection.

	INSPECTION LEVELS					
Lot Size	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	8	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10
35001 to 150000	1250	296	124	74	40	10
150001 to 500000	1250	346	156	91	40	10
500001 and over	1250	431	187	102	40	10

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b. Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

MIL-S-70791 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Nose	of 1	12576516
			NEXT HIGHER ASSEMBLY 12576571

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Length of base, min.	1/, 0-1	3.2	Gage, See Note 1
102	Diameter of spherical end	1/, 0-1	3.2	Gage, See Note 1
Minor				
201	Poor workmanship	1/, 0-1	3.7	Visual, See Note 1, 6.7

NOTES:

1/ See 4.5.6 for qualification and production control of molded parts.

AMSMC Form 1570b, 1 Jul 89 — Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-S-70791 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Tube, Rocket	1 OF 1	12576517
			NEXT HIGHER ASSEMBLY 12576571

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Inside diameter TableII	Level III	3.2	Gage
102	Wall thickness TableII	Level III	3.2	Gage
Minor				
201	Total length TableII	Level IV	3.2	Gage
202	Poor workmanship TableII	Level V	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-S-70791 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Tailcone	1 of 1	12576518
			NEXT HIGHER ASSEMBLY 12576571

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	True position of fin(s) O.D. with bottom	1/, 0-1	3.2	Gage, See Note 1
102	True position of fin(s) thickness	1/, 0-1	3.2	Gage, See Note 1
103	Perpendicularity of bottom	1/, 0-1	3.2	Gage, See Note 1
Minor				
201	Poor workmanship	1/, 0-1	3.7	Visual, See Note 1

NOTES:

1/ See 4.5.6 for qualification and production control of molded parts.

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-S-70791 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Label	1 OF 1	12576520
			NEXT HIGHER ASSEMBLY 12576571

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Marking incorrect or illegible	Table II, Level III	3.2	Visual
Minor 201	Poor workmanship	Table II, Level V	3.7	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-S-70791 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Rocket Assembly (prior to loading)	1 of 1	12576571
			NEXT HIGHER ASSEMBLY 12576572

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	True position of tailcone body	Table II, Level III	3.2	Gage
102	Tube and nose flushness incorrect	Table II, Level III	3.2	Gage
103	Rocket assembly integrity test	15, 0-1	3.5.1	4.5.5/Test
104	Dimensional stability test (Note 1)	15, 0-1	3.5.1.1	4.5.5.1
Minor				
201	Rocket tube flushness incorrect with respect to tailcone	Table II, Level IV	3.2	Gage
202	Poor workmanship	Table II, Level V	3.7	Visual

NOTES:
1/ Samples from Rocket Assembly Integrity test shall be used for the Dimensional Stability Test in the order shown.

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CLASSIFICATION OF CHARACTERISTICS

MIL-S-70791 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Simulator, Missile: sagger XM27	of 1	12576572
			NEXT HIGHER ASSEMBLY: ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Hardness of propellant	5, 0-1	3.2	Dwg. 12576572
102	Propellant present in guide hole	Table II, Level III	3.2	Visual
103	Propellant extends beyond max. limits	Table II, Level III	3.2	Visual/Gage
104	Propellant interferes with guide rod hole	Table II, Level III	3.2	Visual
105	Voids or cracks on propellant surface greater than allowed	Table II, Level III	3.3.1	Visual
106	Propellant separated from walls of rocket (at fin end)	Table II, Level II	3.3.1.1	Visual
107	ID of propellant improper	Table II, Level III	3.2	Gage
108	Function a. Visible Flame b. Rocket hits ground during powered flight c. Range improper	4.4.3.1	3.4	4.5.2
Minor				
201	Decal improperly applied	Table II, Level IV	3.2	Visual
202	Poor workmanship	Table II, Level V	3.7	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Support	1 of 1	12598581
			NEXT HIGHER ASSEMBLY 12598324

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter of hole(s) for tailcone improper	1/, 0-1	3.2	Gage, Note 1
102	Diameter of hole(s) for nosecone improper	1/, 0-1	3.2	Gage, Note 1
103	Angle of mating bevel surface improper	1/, 0-1	3.2	Visual/Gage See Note 1
Minor				
201	Material certification	2/	3.2	Certification
202	Poor workmanship	1/, ASI	3.7	Visual, See Note 1, 6.7
203	Overall length	1/	3.2	Gage, Note 1
204	Overall width	1/	3.2	Gage, Note 1

NOTES: 1/ See 4.5.6 for qualification and production control of molded parts.
2/ With each lot of material submitted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Packing and Marking for Support, Shipping Polystyrene Foam For Simulator, Missile: Sagger XM27	OF 1	12598324
			NEXT HIGHER ASSEMBLY ---

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Incorrect number of simulators	Table II, Level III	3.2	Visual
102	Leakage test on barrier bag	Table II, Level II	3.2	Visual/Leakage Test MIL-P-116
103	Strapping missing, loose or incorrectly assembled	Table II, Level III	3.2	Visual
104	Loose pack	Table II, Level III	3.2	Visual
Minor				
201	Marking incorrect or illegible	Table II, Level IV	3.2	Visual
202	Poor workmanship	Table II, Level V	3.7	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-S-70791 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Box, Pallet Type, Wirebound	1 of 1	12597833
			NEXT HIGHER ASSEMBLY 12598333

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Incorrect number of support boxes	Table II, Level III	3.2	Visual
102	Incorrectly assembled	Table II, Level III	3.2	Visual
103	Loose pack	Table II, Level III	3.2	Visual
Minor				
201	Marking incorrect or illegible	Table II, Level IV	3.2	Visual
202	Poor workmanship	Table II, Level V	3.7	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Function test (hot, cold, ambient). Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria a total of eighty (80) simulator assemblies shall be selected from each lot for test in accordance with 4.5.2 (40 ambient, 20 hot, 20 cold). The lot shall be rejected if two (2) or more major functioning defects occur. After three (3) consecutive lots have met the acceptance criteria above, fifty (50) simulators shall be selected from each lot for test (30 ambient, 10 hot, 10 cold). The lot shall be rejected if one (1) or more major functioning defects occur, and lot sampling shall revert to eighty (80) samples per lot.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Propellant. The propellant shall be tested in accordance with MIL-P-70762. Any lot failing any requirement outlined in the specification and drawings shall be disqualified for use in production.

4.5.2 Function test. (Ambient, 21±3°C (70±5°F), Hot, 51.5±3°C (125±5°F), cold, -32±3°C (-25±5°F)). The simulators shall be conditioned for 12 hours minimum at the specified temperature and fired within 10 minutes after removal from the conditioning chamber. The simulators shall be mounted in an approved firing fixture on a rigid support and fired by means of the specified power source and firing device at an angle of 27.5° ± 2.5° quadrant elevation. Observation shall be made for compliance with the applicable requirements. Any rocket that fails to comply with requirements shall be classified as a defect. NOTE: Rocket shall not be tested in wind conditions of 15 knots head wind or 15 knots crosswinds.

4.5.3 Transportation vibration - temperature. Simulator samples shall be tested according to Test 119 MIL-STD-331, Procedure I. Five (5) rockets shall be pre-conditioned at 0°F for two (2) hours max. prior to vibration and the other five (5) rockets at 145°F for two (2) hours max. prior to vibration. The vibration test duration at each temperature shall be six (6) hours (2 hours per axis). Simulators shall be functioned at ambient temperature in accordance with 4.5.2. (First Article Only)

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4.5.4 Five foot drop. The rocket shall be tested in accordance with MIL-STD-331, Test 111.1, Procedure 2. Ten (10) rockets shall be dropped at ambient temperature in each of the five orientations required in the procedure. Observation shall be made for compliance with applicable requirements. The rockets shall be function tested after drop per 4.5.2. Any rocket that fails to function shall be classified defective (First Article only).

4.5.5 Rocket assembly integrity test. The rocket assembly (dwg. 12576571) shall be subjected to an internal pressure of 40-45 psig for 15 sec min. to verify requirements specified in 3.5.1. The rocket assembly shall be restrained by the tailcone only for this test. Any assembly which fails to comply with the applicable requirements of 3.5.1 shall be classed defective (Non-destructive test).

4.5.5.1 Rocket assembly dimensional stability test. Prior to the test, the following dimensions are to be measured:

a) Nose diameter at spherical end
b) Tailcone guide rod hole diameter, min
c) Fin(s) outside diameter
d) Perpendicularity of tailcone bottom
e) Overall length

The rocket assembly (dwg. 12576571) shall be subjected to a temperature of 135 $\pm$ 5°F for 24 hours minimum with the assembly in vertical position, nose end down resting on a solid surface with an 8 pound weight applied to the tail end. After assembly is allowed to reach ambient temperature, the above dimensions shall be rechecked for conformance with the applicable requirements (of less than 2% in variation) specified in 3.5.1.1. Any assembly which fails to comply with the applicable requirements shall be classed defective (Non-destructive test.)

4.5.6 Dimensional control of molded or plastic parts. In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (See Note 1), a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally (all dimensions on drawing) to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in the pertaining paragraphs 4.4.2.1, 4.4.2.3, and 4.4.2.7 after

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continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one(1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective parts will not be used in production. If any defective part(s) are found when inspection is performed for the control of the cavity, the cavity producing the defective part(s) shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for appropriate action, in accordance with the provisions of the contract. All cavities removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Contractor designs of gages and test equipment required to perform the inspections listed herein shall be forwarded to the technical agency specified in contract for approval prior to manufacture of equipment. The noted sub-paragraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1: In establashing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60 etc. minutes) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to the technical agency specified in contract. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

5. PACKAGING

5.1 Packaging requirements for missile sagger simulator.

5.1.1 Packaging.

5.1.1.1 Level A. Not applicable

5.1.1.2 Level B. Packaging shall be in accordance with Dwg. 12598324.

5.1.1.3 Level C. Same as level B.

5.1.2 Packing.

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5.1.2.1 Level A. Not applicable.

5.1.2.2 Level B. Packing shall be in accordance with Dwg. 12598333.

5.1.2.3 Level C. Same as level B.

5.1.3 Marking. Marking shall be in accordance with Dwgs. 12598324 and 12598333.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for use on the Simulator, Missile: sagger XM27.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the Contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Distribution of ammunition data cards. Distribution of data cards include the following: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-M (D), Picatinny Arsenal, NJ 07806-5000.

6.5 Humidity control. Pyrotechnic compositions are hygroscopic materials and will deteriorate rapidly when exposed to moisture. It is highly desirable that all the ingredients of the simulator be carefully safe guarded from exposure to moisture and humid atmosphere. For assurance of the satisfactory loading of the simulator it is recommended that the blending and loading building be air conditioned in a manner that will maintain the relative humidity of the atmosphere between 30% and 60%.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings, prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.7 Subject term (key word) listing.
Missile sagger Simulator
Propellant
Rocket Range
Visible Flame

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1370-A284)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-70791 (AR)	2. DOCUMENT DATE (YYMMDD) 9 APRIL 1990

3. DOCUMENT TITLE
SIMULATOR, MISSILE: SAGGER XM27, PARTS FOR AND LOADING, ASSEMBLING AND PACKING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US ARMY ARDEC STANDARDIZATION & SPECIFICATION OFFICE	b. TELEPHONE (Include Area Code) (1) Commercial: (201) 724-6671 (2) AUTOVON: 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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